Chapter 1: Introduction

Introduction

- Unlocking the Power of Databases
- Imagine a world without databases:
  - your favorite social media platforms wouldn’t remember your login,
  - banks wouldn’t track your transactions, and
  - online shopping would be impossible.
- Databases are the invisible engines powering our digital world, ensuring data is stored, accessed, and managed efficiently.

Definitions

- Database (DB): an organized collection of data
- Database Management System (DBMS): a group of programs to:
  - Manages and Manipulates database
  - Provides an interface between the database and the user and other application programs
  - Contains information about a particular enterprise
  - An environment that is both convenient and efficient to use
- Database Administrator (DBA): a skilled professional who directs all activities related to an organization's database
- Database Systems process and organize large amounts of data (Ex: Bank accounts database system)

Data Management

- Without data and the ability to process it, an organization could not successfully complete most business activities
- For data to be transformed into useful information, it must first be organized in a meaningful way
- Examples: MySQL (powering Facebook’s backend), Oracle (used in global banking networks), and PostgreSQL (supporting Instagram's growing data needs).

Features of DBMS

1. Data storage, retrieval, and update: The ability to store, retrieve, and update the data in the database.
2. User-accessible catalog: where database component descriptions are stored and accessible to the users.
3. Shared update support: A mechanism to ensure accuracy when several users are updating the database at the same time.
4. Backup and Recovery Services: Mechanisms for recovering the database if a database is damaged somehow.
5. Security Services: Mechanisms to ensure that certain rules are followed about data in the database and any changes that are made in the data.

Features of DBMS

6. Integrity services: Mechanisms to ensure that certain rules are followed about data in the database and any changes made in the data.
7. Data Independence: Facilities to support the independence of programs from the structure of the database.
8. Replication support: A facility to manage copies of the same data at multiple locations.
9. Utility Services: DBMS provides services that assist in the general maintenance of the database.

Main Functions

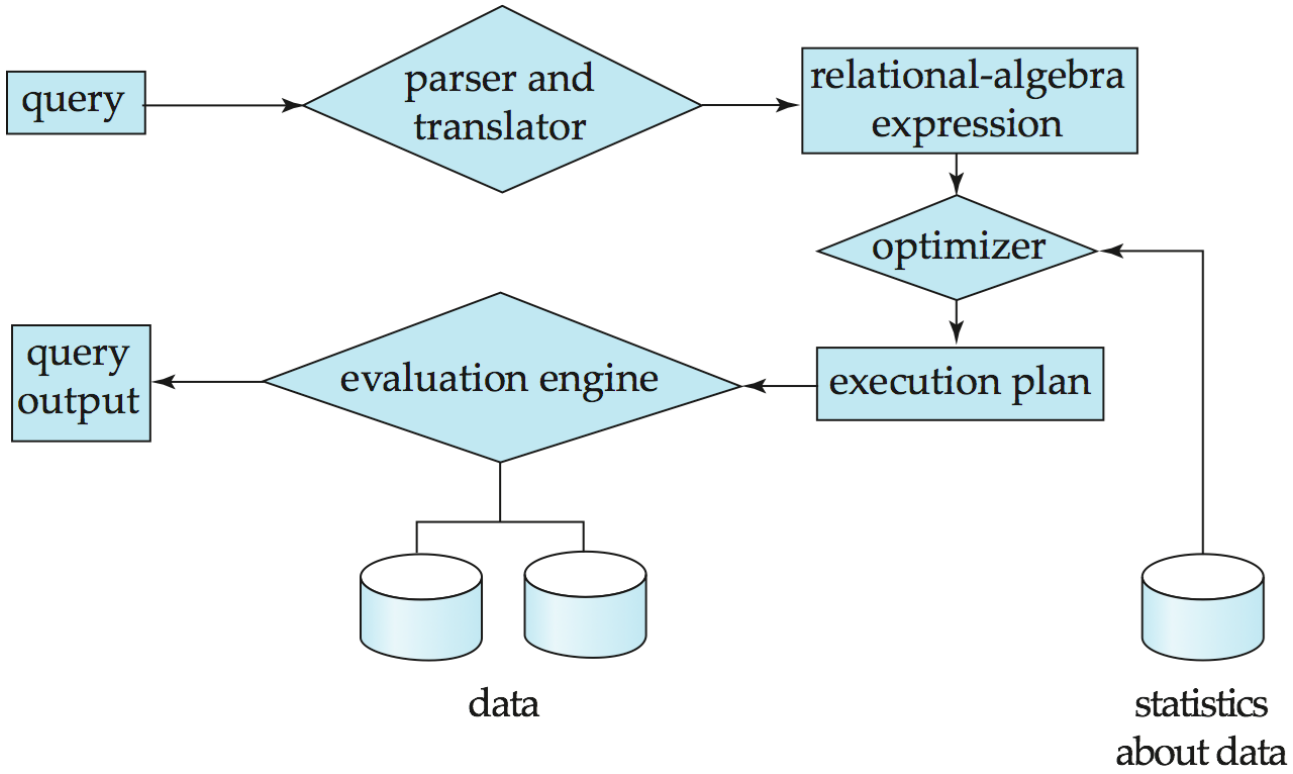
1. Storage manager
2. Query processing
3. Transaction manager

1. Storage Management

- Provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.
- The storage manager module is responsible for the following tasks:
  - Interaction with the OS file manager.
  - Efficient storing, retrieving, and updating of data.
- Issues:
  - Storage Access
  - File organization
  - Indexing and hashing

2. Query Processing

1. Parsing and translation
2. Optimization
3. Evaluation



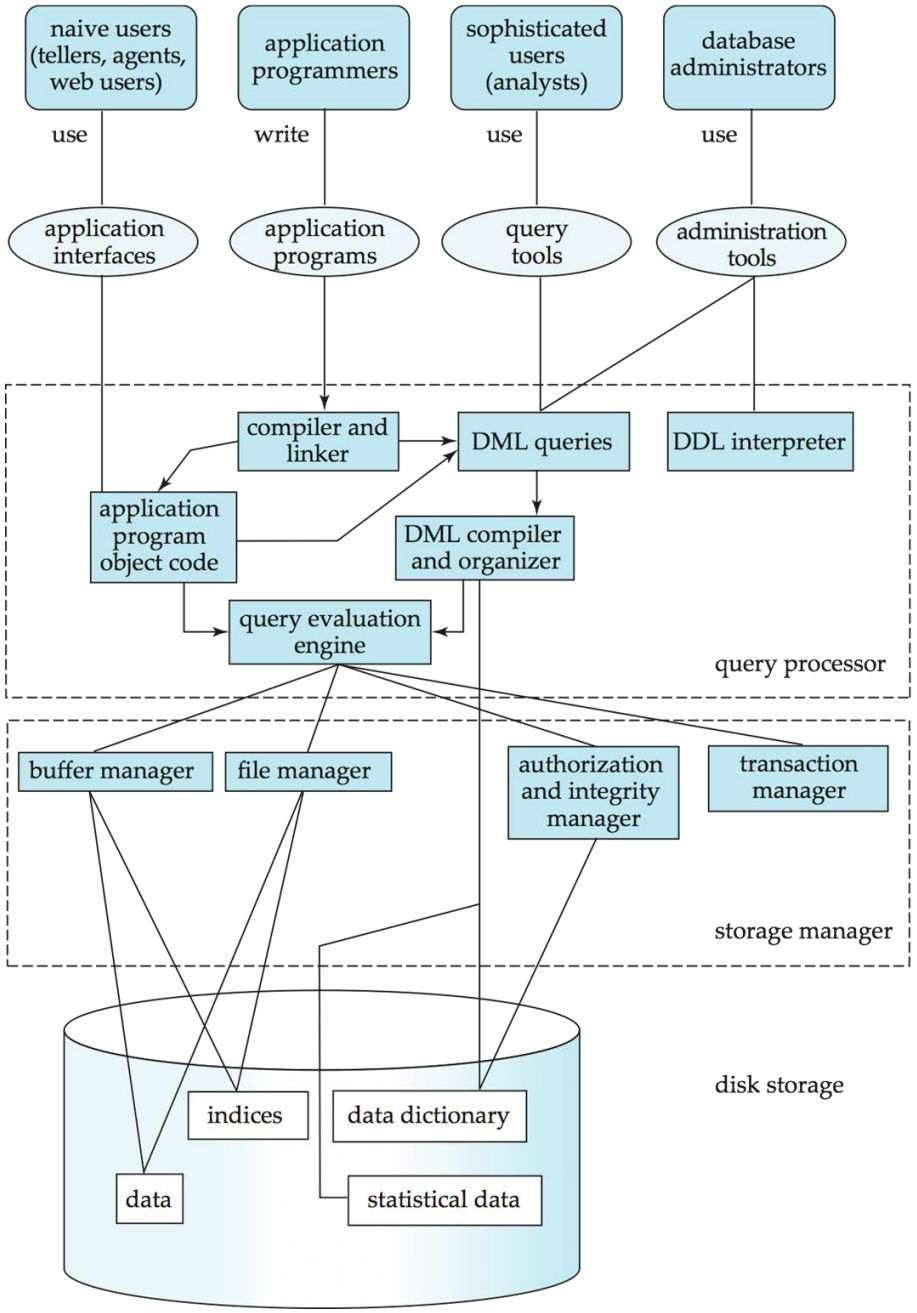
Query Processing (Cont.)

- Alternative ways of evaluating a given query
  - Equivalent expressions
  - Different algorithms for each operation
- Cost difference between a good and a bad way of evaluating a query can be enormous
- Need to estimate the cost of operations
  - Depends critically on statistical information about relations which the database must maintain
  - Need to estimate statistics for intermediate results to compute cost of complex expressions

3. Transaction Management

- What if the system fails?
- What if more than one user is concurrently updating the same data?
- A transaction is a collection of operations that performs a single logical function in a database application
- Transaction-management component ensures that the database remains consistent (correct) despite system failures (e.g., power failures and operating system crashes) and transaction failures.
- Concurrency-control manager controls the interaction among the concurrent transactions to ensure the consistency of the database.

Database System Internals



Database Architecture

The architecture of a database system is greatly influenced by the underlying computer system on which the database is running:

- Centralized
- Client-server
- Parallel (multi-processor)
- Distributed

Popular DBMS

- Popular DBMSs for end users: Microsoft Access and FileMaker Pro
- Entire market includes databases by IBM, Oracle, and Microsoft
- Examples of open-source database systems: PostgreSQL and MySQL
- Many traditional database programs are now available on open-source operating systems

The Titans of Database Technology

- Enterprise-Level: Oracle (global banking), Microsoft SQL Server (corporate data management), IBM Db2 (financial institutions).
- Open Source: MySQL (WordPress sites), PostgreSQL (Instagram’s backend), MariaDB (Wikipedia’s content storage).
- Cloud-Based: Google Cloud Spanner (YouTube video metadata), Amazon RDS (startups & enterprises), Microsoft Azure SQL (cloud applications).

Selecting a DBMS

- Important characteristics of databases to consider
  - Size of the database
  - Cost of the system
  - Number of concurrent users
  - Performance
  - Ability to be integrated with other systems
  - Vendor considerations

Selecting a DBMS

- Size Matters: SQLite for mobile apps vs. Oracle for handling massive business data.
- Budget Concerns: Open-source alternatives vs. premium services like SAP HANA.
- Performance & Scale: MongoDB (eBay's dynamic product listings) thrives in handling unstructured data.
- Integration: Essential for seamless connectivity with ERP (SAP), CRM (Salesforce), and cloud platforms (AWS, Google Cloud).

Using DBMS with Other Software

- DBMS are often used with other software packages or the Internet
- A DBMS can act as a front-end application or a back-end application
  - Front-end application: interacts with users
  - Back-end application: interacts with applications